

Element C: Presentation and Justification of Solution Design Requirements

Dogs are getting into indoor trash cans, making messes, creating health risks, and causing inconvenience for dog owners and people inside the home. This problem involves dog owners, the dogs themselves, and other people inside the home. This happens inside the house nearly every day. Dogs dig in the trash because they are attracted to the food and smells inside of the trash cans. This is a small but persistent issue that leaves owners irritated.

We have nine design requirements that we will try to follow throughout this project. If we follow these design requirements, we will develop a better product and our stakeholders will be satisfied. If we meet the requirements our product should address all the trash can problems for any dog owner with any type of dog.

1. Prevents dogs from getting to/ripping up trash bags (High)
2. Does not harm dogs or owners (High)
3. Durable enough to prevent dogs from breaking the device (High)
4. Universal for all trash cans (High)
5. Works for dogs all sizes (High)
6. Must cost \$20 or less to make (High)
7. One hand is able to operate trash can (Medium)
8. Does not require specialty tools to install (Medium)
9. Installs in 10 minutes or less (Medium)
10. Available in multiple colors (Low)

Our high priority design requirements are the most important, and we should address them all. These are the most important because they will address the major

issues of our problem. Our stakeholders would be much happier if we met all of them and missed a few low or medium priorities, than if we met all medium and low and missed two high priority requirements.

We need to prevent the dogs from getting to the trash bags otherwise this project would be a waste. We must not harm dogs or their owners because we value safety. We must make our final solution relatively cheap because we realize that many people are on a limited budget. The next solution must be durable enough to prevent dogs from breaking it because it cannot do much if it is broken. The next solution must work against dogs of all sizes because so any dog owner can use it. The final solution is it must be universal for all trash cans so people with any type of trash can use it. All of these design requirements are necessary to build a solution that is functional, easy to install, and meets all of our stakeholders' needs.

One of our stakeholders Mrs. L mostly agreed with our design requirements, and agreed with our cause. They said that it would help keep dogs safe, and saved themselves from having to clean up the mess their dogs make. One of our stakeholders said that they would gladly pay over \$20 to buy our product if it was an actual product that worked really well. According to another one of our stakeholders named C, he believed that our proposed solution seemed like a great foolproof solution to the problem of dogs getting into trash. C also said that he would also pay \$20 or more for such a product if it is proven to work. He suggested that when manufacturing this product we could also branch out to provide other barriers to keep dogs out of places they shouldn't be, like on furniture or stealing food from counters.